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The Energy Costs of Cooling Electronic Systems Reflections on Past Lessons and Future Opportunities

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Courtesy of Steel Orca

Acknowledgements

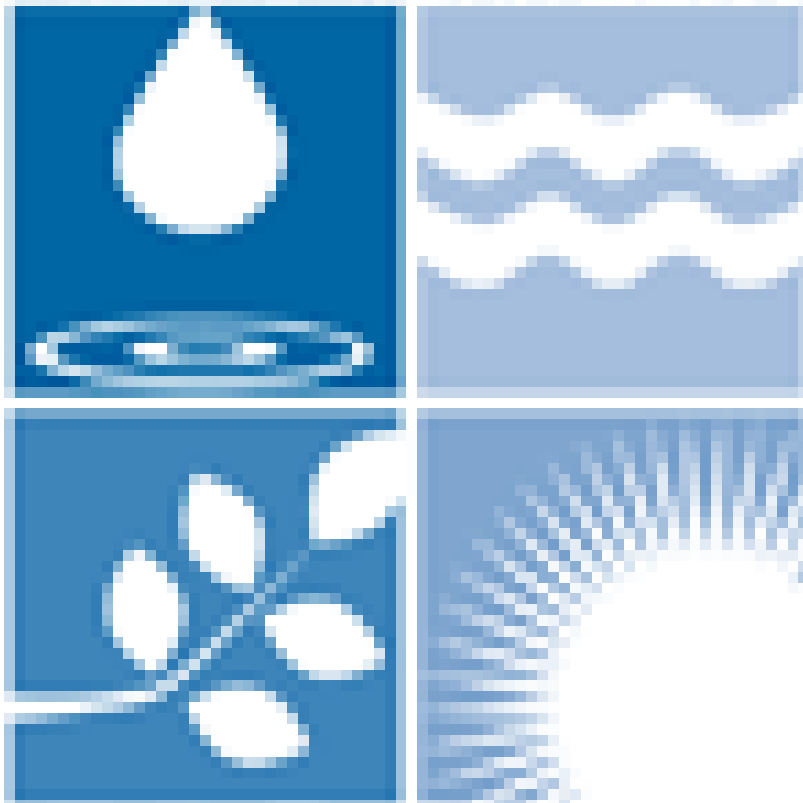
www.binghamton.edu/e3s



E3S

Center for Energy-Efficient
Electronic Systems

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Discussion Points

SUSTAINABLE DESIGN PRACTICES

ENERGY EFFICIENCY: REDUCING ENERGY COSTS BY PURPOSEFUL IDENTIFICATION OF IRREVERSIBILITIES

THE ENERGY COST OF THERMAL DESIGN DRIVEN BY PERFORMANCE PER UNIT COST (PUC)



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THE ENERGY COST OF THERMAL DESIGN DRIVEN BY PERFORMANCE PER UNIT COST (PUC)

**MOST ELECTRONICS COOLING ENGINEERS
HAVE SPENT CAREERS SEEKING THERMAL
DESIGNS THAT SATISFY PERFORMANCE
CRITERIA PER UNIT COST**

**THE ENERGY COSTS OF THESE SOLUTIONS
HAVE NOT BEEN DESIGN DRIVERS**



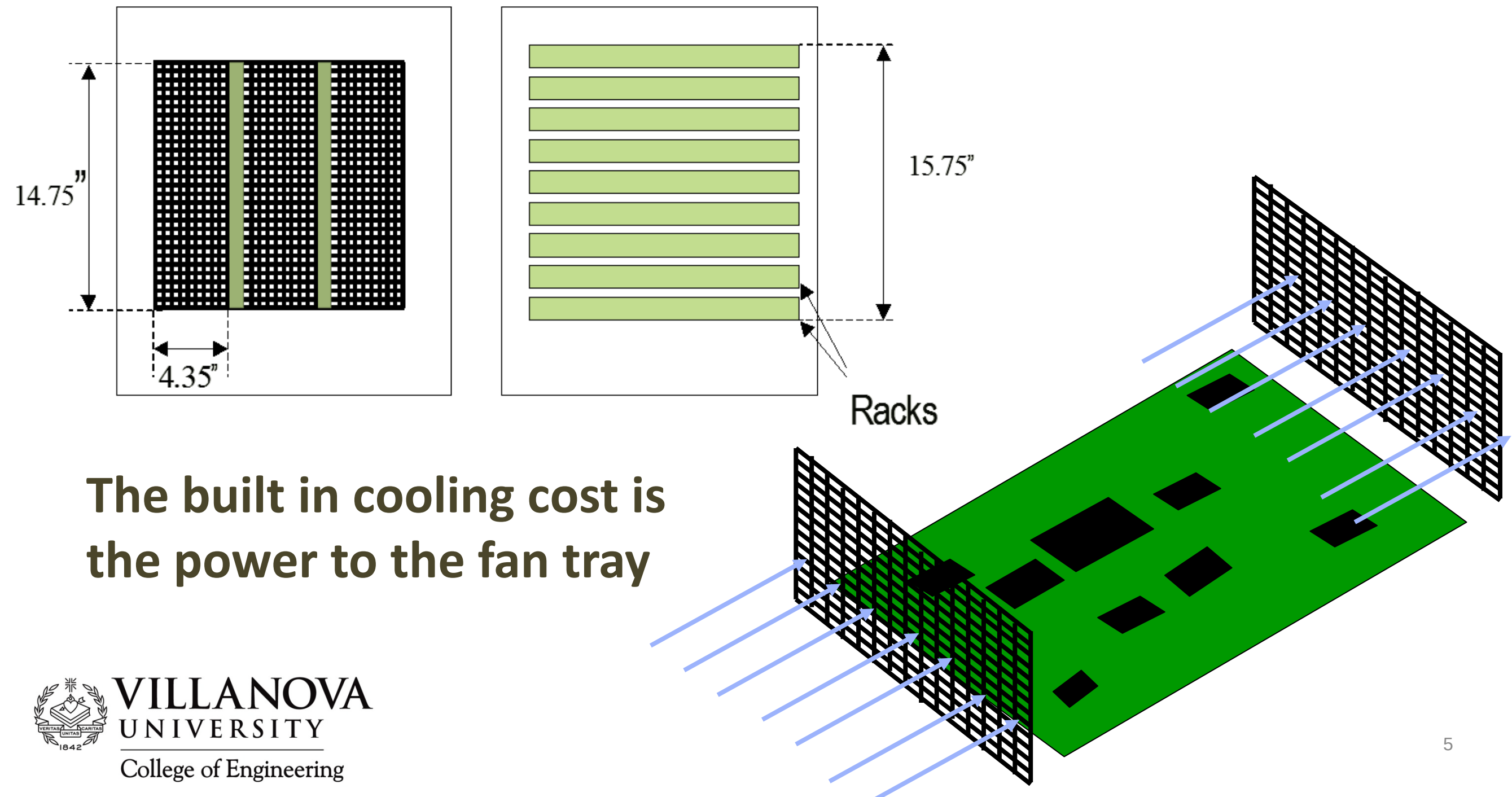
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COOLING PROBLEM: REMOVING HEAT EFFECTIVELY FROM DISTRIBUTED SOURCES

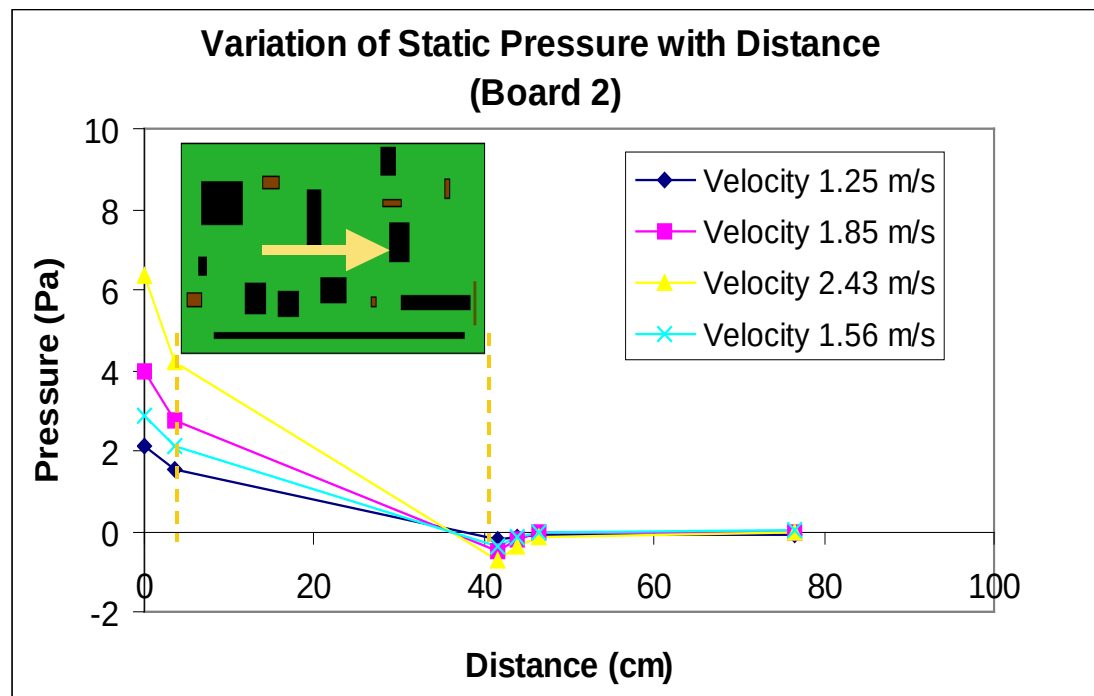
Side View

Front View



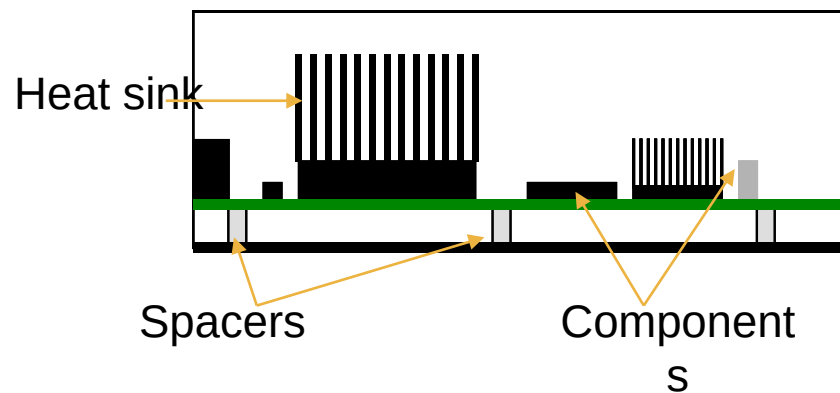
**The built in cooling cost is
the power to the fan tray**

SYSTEM HEAD LOSS

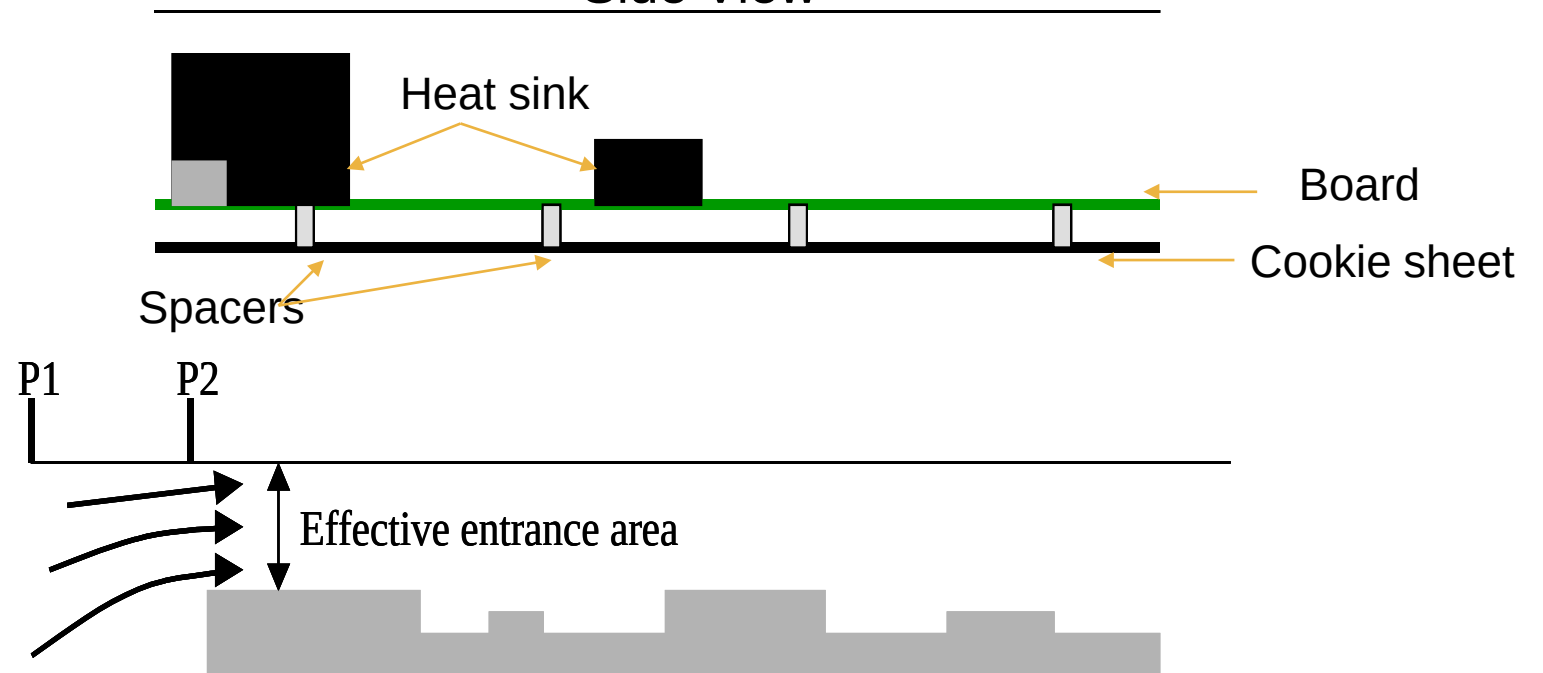


Pressure drop across grills
and boards must be
matched by fan

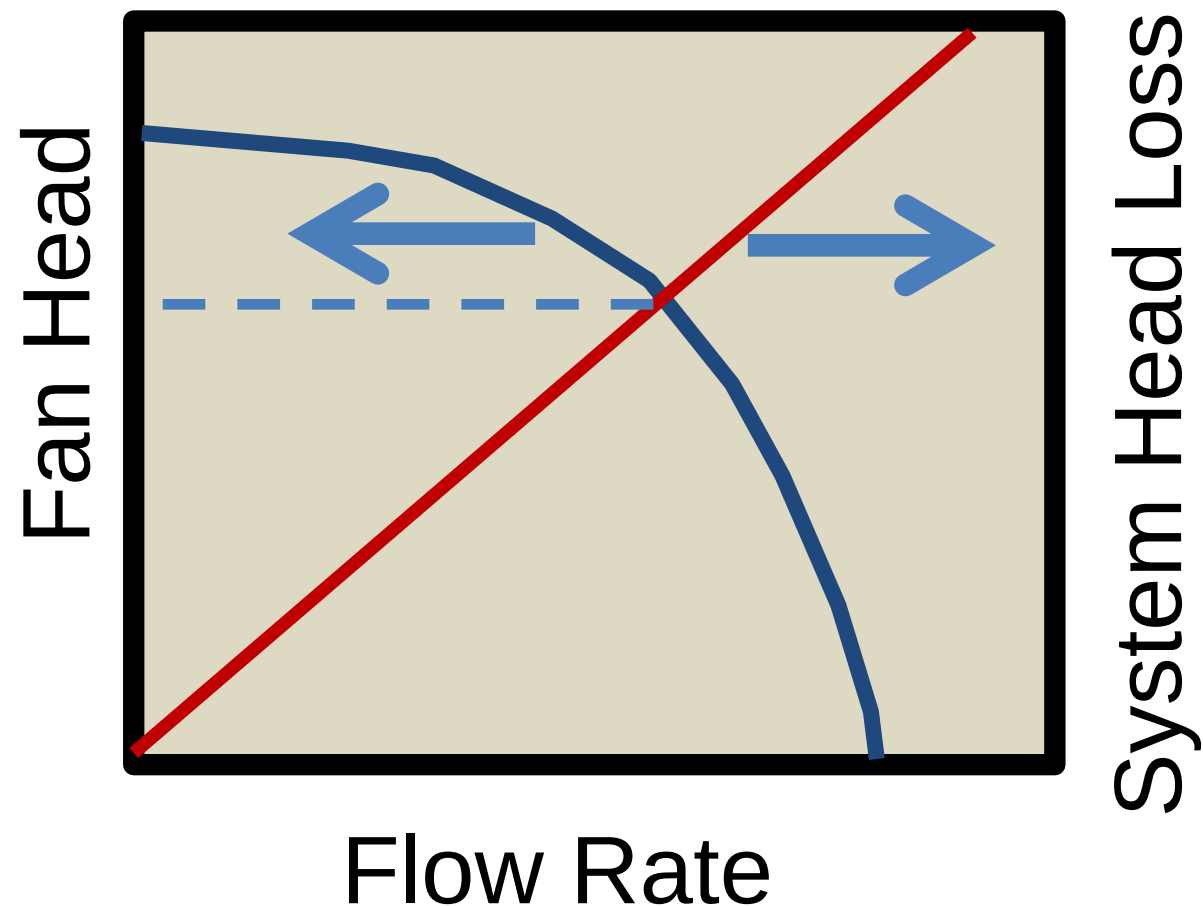
Front View



Side View



THE ENERGY COST OF FAN AIR FLOW



$$\dot{P}_{blower} = \dot{m} \left(\frac{\Delta P}{\rho g} \right)$$

The blower work is the energy cost of producing flow for cooling

ENTROPY GENERATION AND EXERGY DESTRUCTION

First Law:

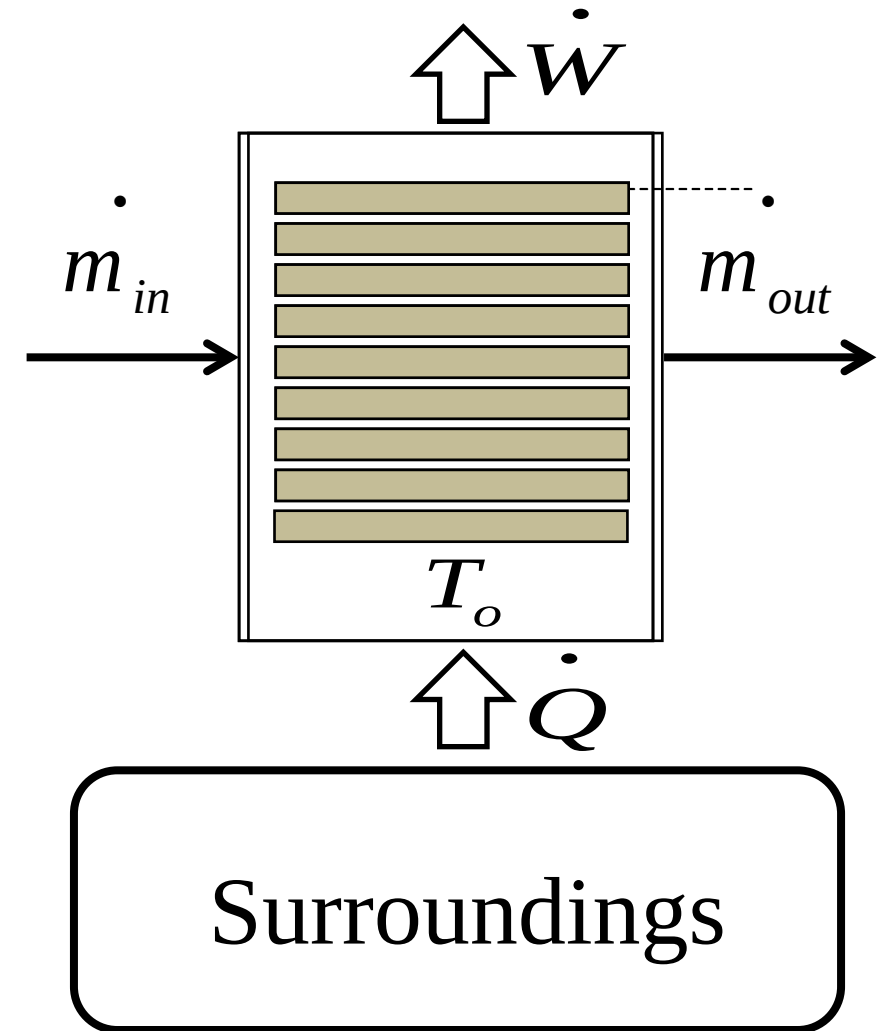
$$\dot{Q} - \dot{W} = \dot{m}_{out} h_{out} - \dot{m}_{in} h_{in}$$

Second Law:

$$\dot{S}_{gen} + \frac{\dot{Q}}{T_o} + \dot{m}_{in} s_{in} = \dot{m} s_{out}$$

Combine the two Laws:

$$\dot{W} = \dot{m}_{in} (h - T_o s)_{in} - \dot{m}_{out} (h - T_o s)_{out} - T_o \dot{S}_{gen}$$



ENTROPY GENERATION AND EXERGY DESTRUCTION

Since there is no attempt to recover any work

$$T_o \dot{S}_{gen} = \dot{m}_{in} (h - T_o s)_{in} - \dot{m}_{out} (h - T_o s)_{out}$$

or

$$T_o \dot{S}_{gen} = \dot{m}(b_{in} - b_{out}) \quad \longrightarrow \quad \text{Lost Available Work}$$

and if there is no heat transfer, we can immediately show

$$T_o \dot{S}_{gen} = \dot{m} \left(\frac{\Delta P}{\rho g} \right)$$

Lost Available Work of course is equal to the Blower Work because all of the Blower work is consumed by the pressure drop irreversibilities



SUMMARY

ALL REAL THERMAL FLUID DESIGNS HAVE IRREVERSIBILITIES OR LOSSES

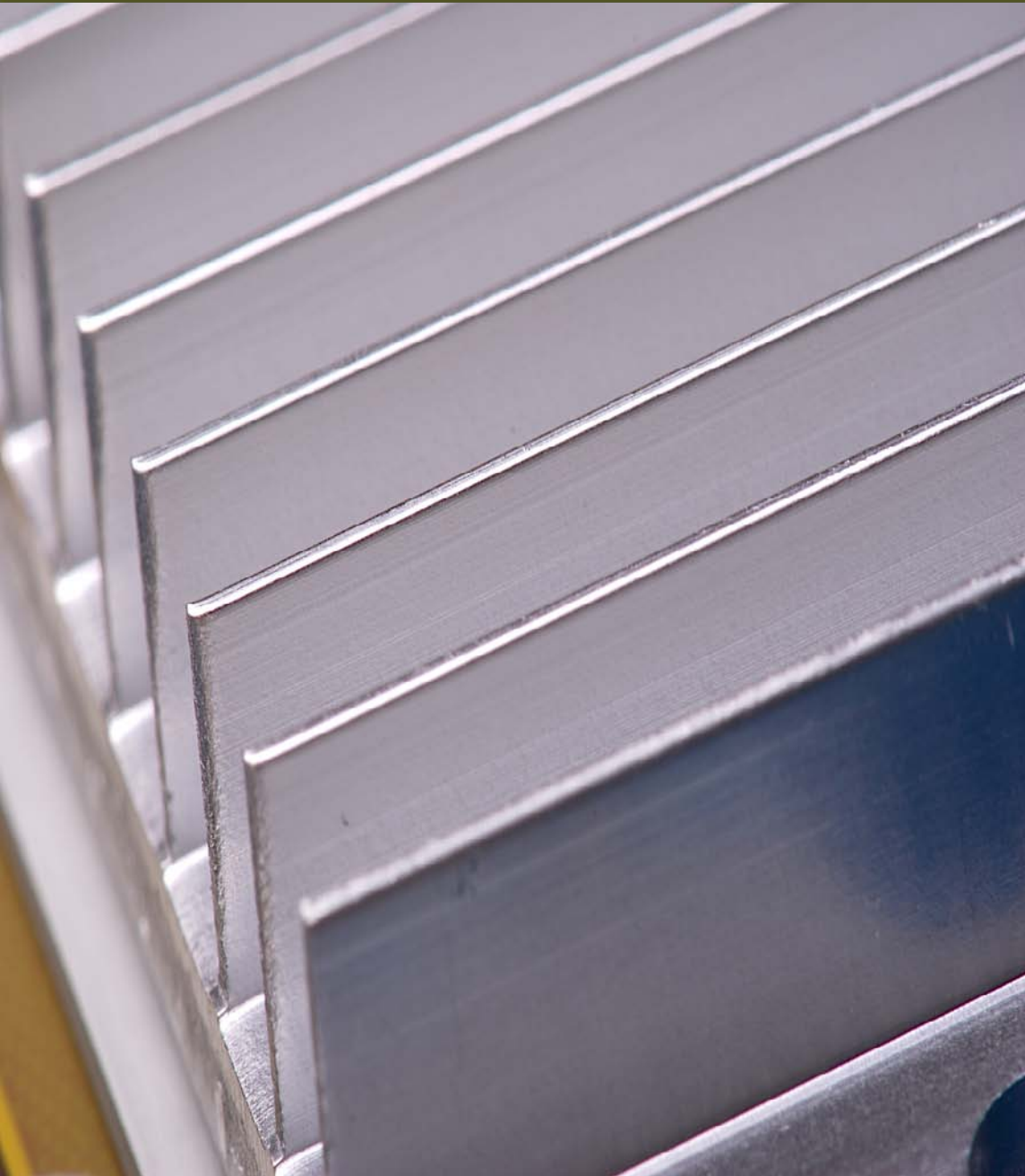
THESE LOSSES LEAD TO THE DESTRUCTION OF AVAILABLE WORK (NOW MORE COMMONLY CALLED “EXERGY”)

$T_o \dot{S}_{gen}$ REPRESENTS THE RATE AT WHICH AVAILABLE WORK OR EXERGY IS DESTROYED

IT REPRESENTS THE ENERGY COST OF INEFFICIENCIES



THE ENERGY COSTS OF COST DRIVEN HEAT EXCHANGERS



Using air cooling through extruded aluminum parallel plate heat sinks is an example of extreme cost driven thermal design

We can show that by common measures used to rate heat exchangers, these types of heat sinks have inferior effectiveness

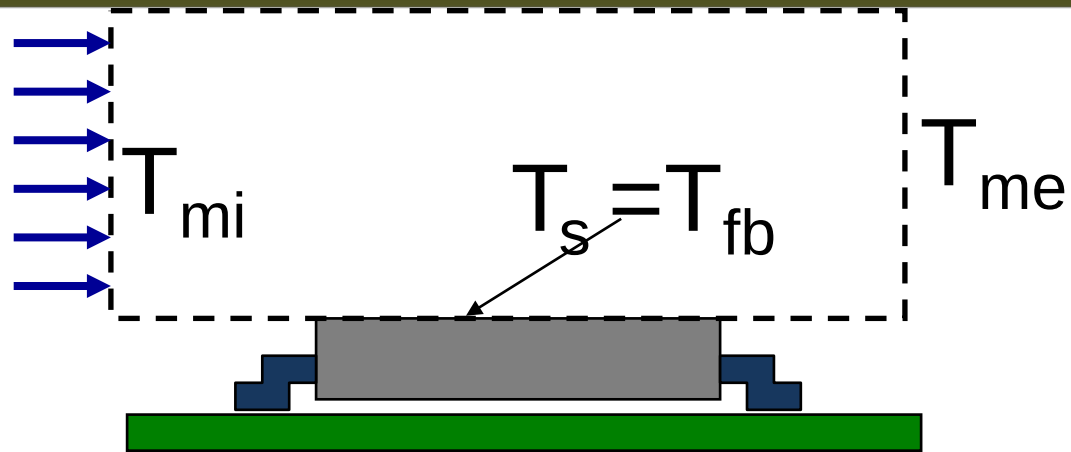
This ineffectiveness leads to high Exergy Destruction and therefore excessive energy costs



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Adapting Heat Exchanger Effectiveness to a Heat Sink or Cold Plate



$$R_{b-a} = \frac{T_{fb} - T_{m,i}}{Q_{total}} = \frac{(T_{fb} - T_{m,i})}{\dot{m} C_p (T_{m,o} - T_{m,i})}$$

Here we assume that T_{fb} is the uniform temperature of the base of the fins

Therefore

$$R_{b-a} = \frac{1}{\dot{m} C_p \varepsilon} = \frac{1}{\dot{m} C_p \left[1 - e^{-\frac{\eta_{o(c)} h A_t}{\dot{m} C_p}} \right]}$$

Note that because the fin surfaces are not isothermal, we introduce the fin efficiency, $\eta_{o,c}$

Note that $NTU = \frac{\eta_{o(c)} h A_t}{\dot{m} C_p}$

Interpretation of ε and NTU

$$\varepsilon = \frac{(T_{m,o} - T_{m,i})}{(T_{fb} - T_{m,i})}$$

Its maximum value is 1.0 when

$$T_{m,o} = T_{f,b}$$

i.e. when the fluid exits at its maximum possible temperature which is the temperature of the base of the fin, thus

$$R_{b-a,\min} = \frac{1}{\dot{m} C_p} \quad \text{Thermodynamic Limit}$$

$$\frac{R_{b-a}}{R_{b-a,\min}} = \frac{1}{\varepsilon} = \frac{1}{[1 - e^{-NTU}]}$$

$$NTU = \frac{\left(\frac{1}{\dot{m} C_p} \right)}{\left(\frac{1}{\eta_{o(c)} h A_t} \right)} = \frac{R_{advective}}{R_{fins+convective}}$$

For small NTU, $\dot{m} C_p \rightarrow \infty$ or $\eta_{o(c)} h A_t \rightarrow 0$
 $\varepsilon \approx NTU$ and therefore $R_{b-a} \rightarrow \frac{1}{\eta_{o(c)} h A_t}$

For large NTU, $\dot{m} C_p \rightarrow 0$ or $\eta_{o(c)} h A_t \rightarrow \infty$
 $\varepsilon \rightarrow 1.0$ and $R_{b-a} \rightarrow R_{b-a,\min} = \frac{1}{\dot{m} C_p}$

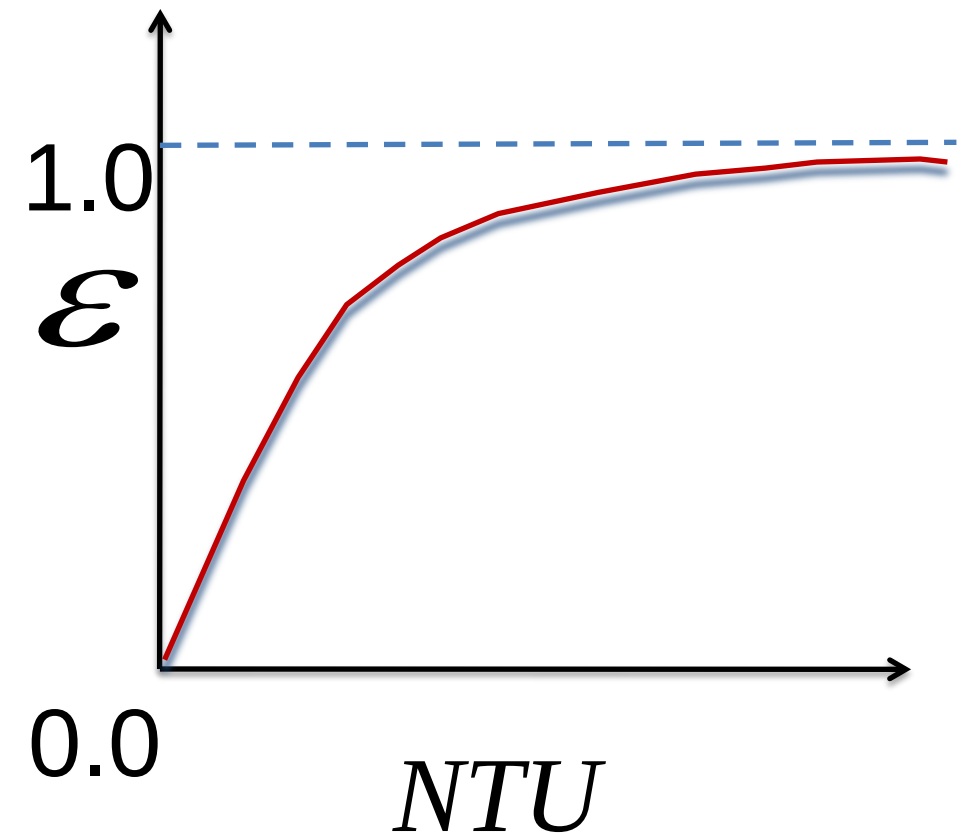
Effective means achieving design R with least flow rate

$$R_{b-a} = \frac{T_{fb} - T_a}{Q_{total}} = \frac{1}{\dot{m} C_p \varepsilon}$$

The lower the effectiveness, the more mass flow required to achieve R_{b-a}

To achieve high values of effectiveness we need to achieve high values of NTU but not by using excessive mass flow rate

$$\eta_{o(c)} h A_t \rightarrow \text{LARGE}$$

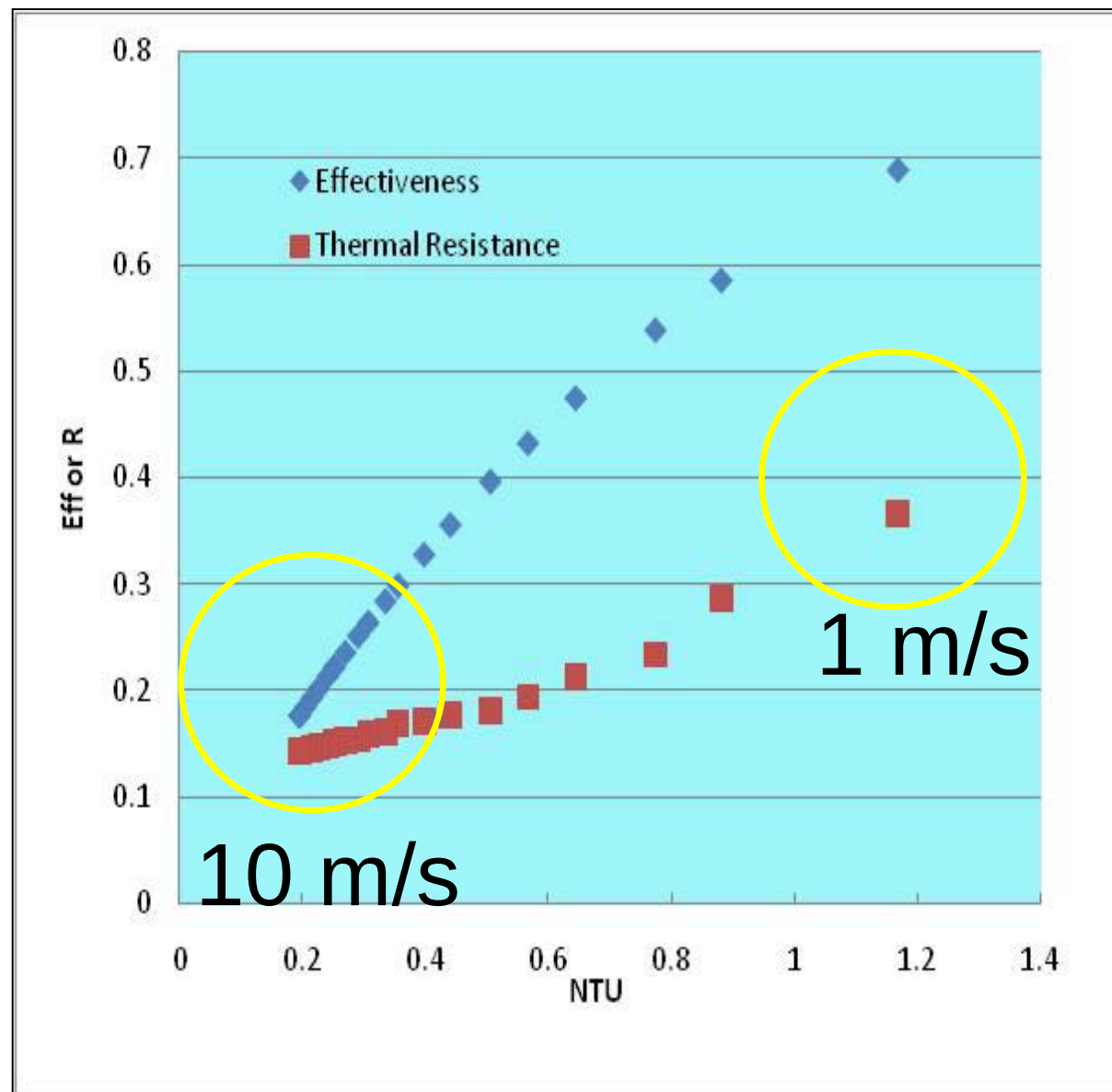


$$NTU = \frac{\left(\frac{1}{\dot{m} C_p} \right)}{\left(\frac{1}{\eta_{o(c)} h A_t} \right)} = \frac{R_{advective}}{R_{fins+convective}}$$

Air-cooling has severe limitations

Air-cooled Cu-Cu 1.5 mm pitch—a “high performance heat sink”

- Effectiveness and R



- For a desired R value of 0.15
- The heat sink effectiveness is 0.18
- Implications: to achieve the required R value, we require too much flow rate of air
- This cost driven solution has hidden energy costs

SUMMARY

AIR COOLING WITH COMMON HEAT SINKS IS A COST DRIVEN THERMAL MANAGEMENT SOLUTION

DESPITE ITS “LOW COST” IT HAS HIDDEN ENERGY COSTS

**MEETING TARGET PERFORMANCE REQUIREMENTS
REQUIRES EXCESSIVE FLOW RATES AND PRESSURE DROP**

**AND CONSEQUENTLY, STRONGER IRREVERSIBILITIES
AND HIGHER RATES OF EXERGY DESTRUCTION**

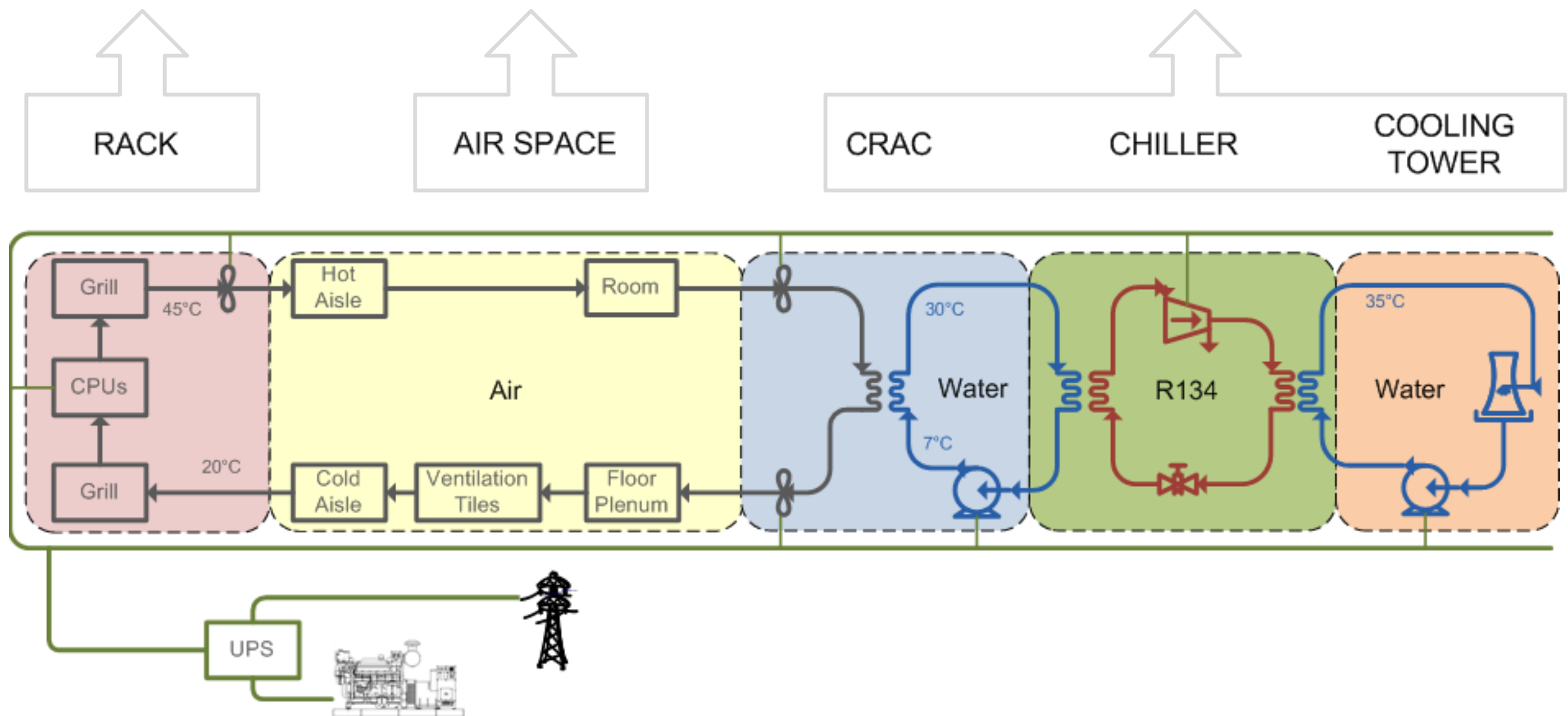


APPLYING SUSTAINABLE ENGINEERING PRACTICES IN DATA CENTERS

- **POWER**
 - Eliminating wasteful AC/DC conversion steps
 - Alternatives to grid power
- **EFFICIENCY**
 - Improving Legacy Air-Cooled designs by identifying sources of irreversibilities
- **WASTE ENERGY RECOVERY**
 - Reusing energy that is normally wasted



ENERGY EFFICIENCY: REDUCING ENERGY COSTS BY PURPOSEFUL IDENTIFICATION OF IRREVERSIBILITIES



* Source: Tony Evans, *The Different Types of Air Conditioning Equipment for IT Environments*, White Paper #59, American Power Conversion.

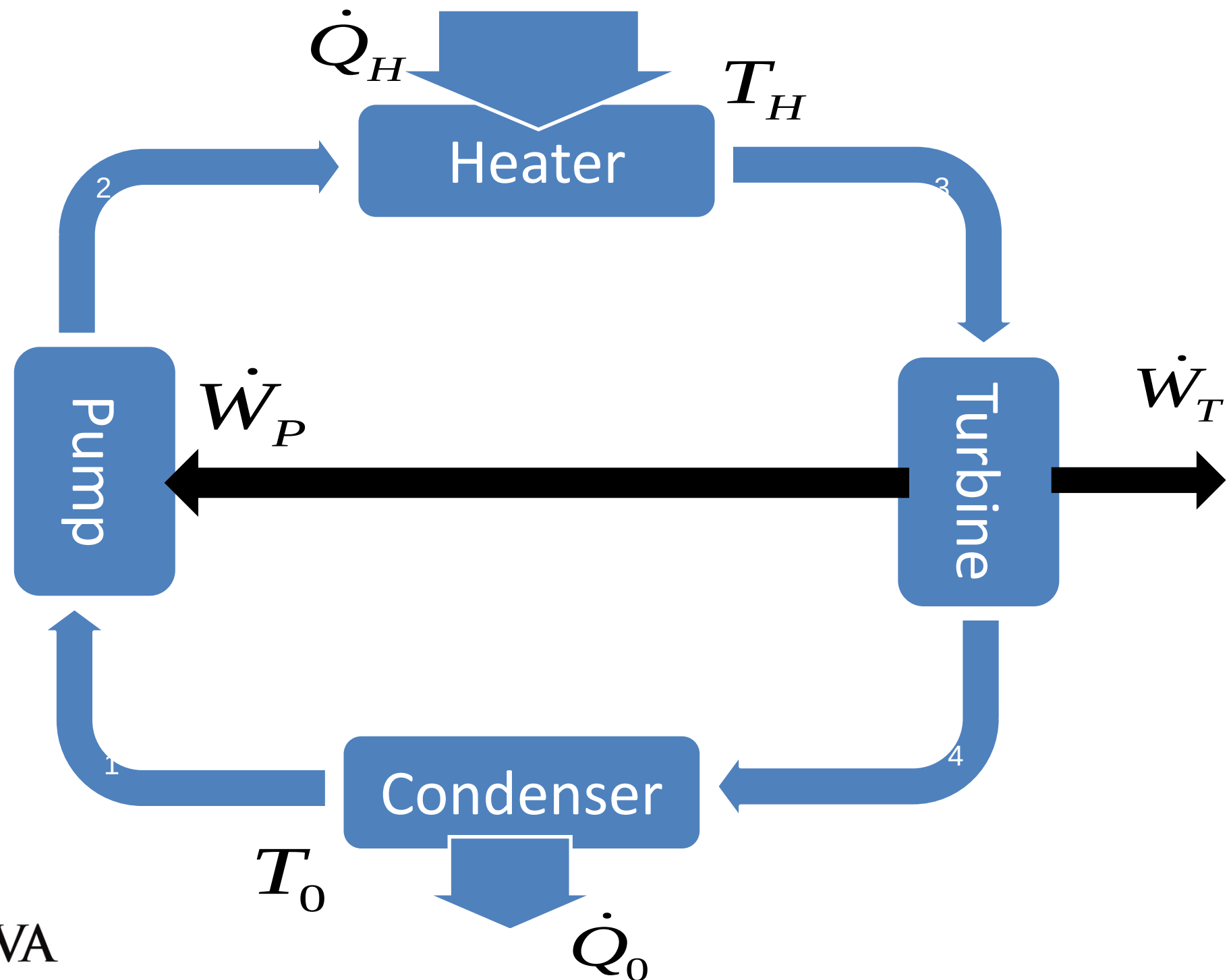
** Source: <http://www.mentor.com/company/news/apc-uses-airflow-simulation-to-solve-data-center-cooling-problem>



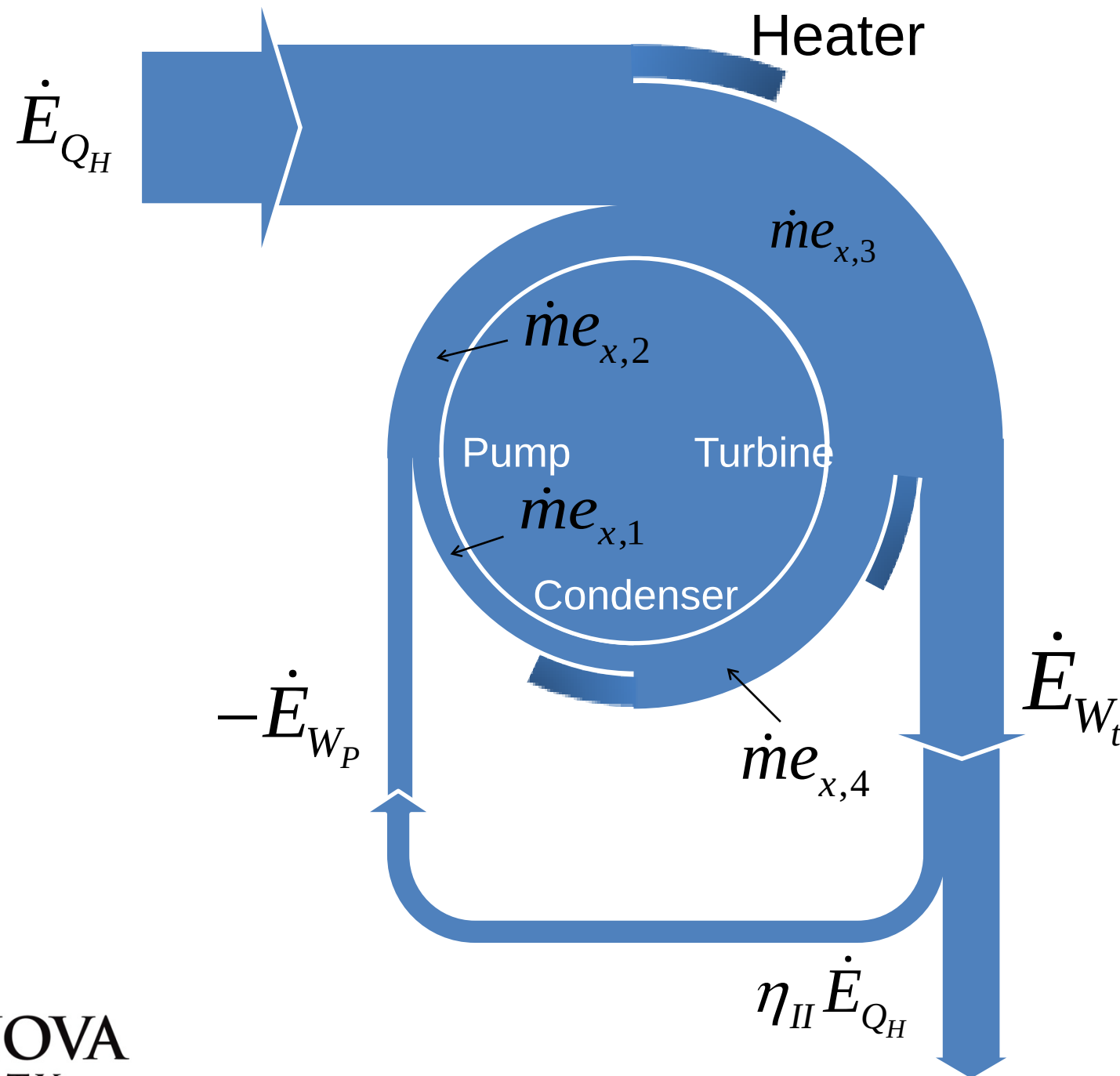
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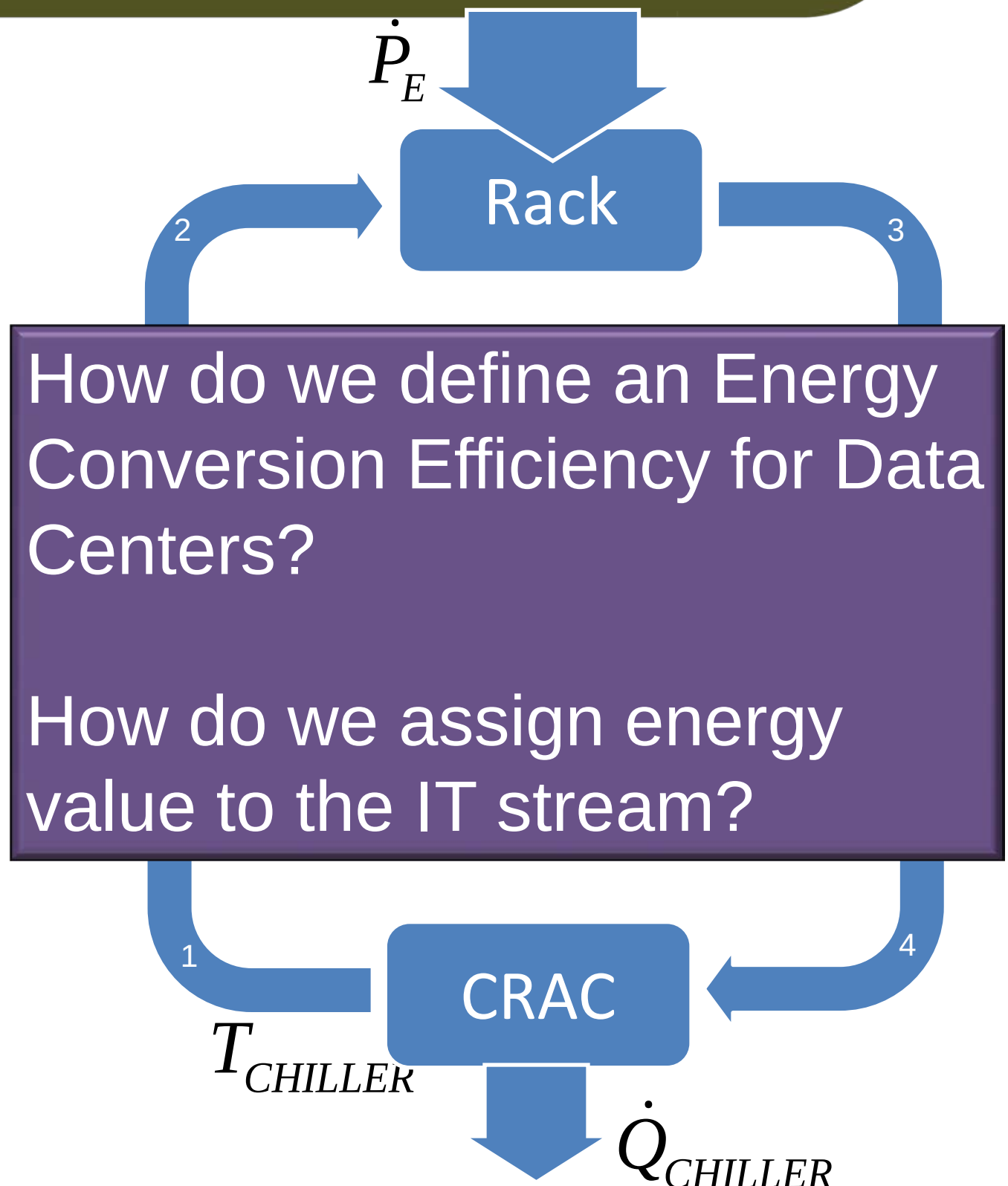
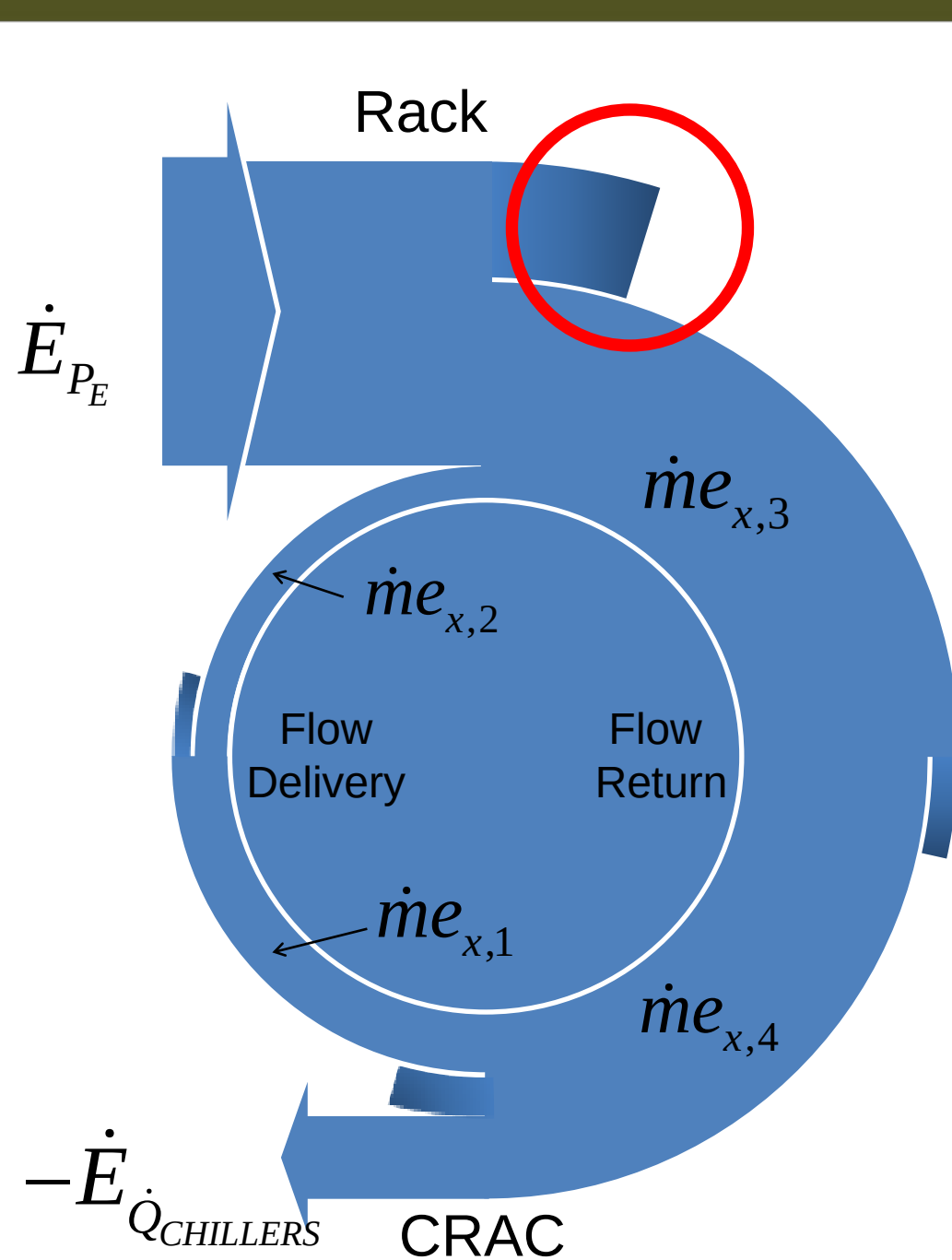
Rankine Power Cycle



Exergy Destruction Rankine Cycle

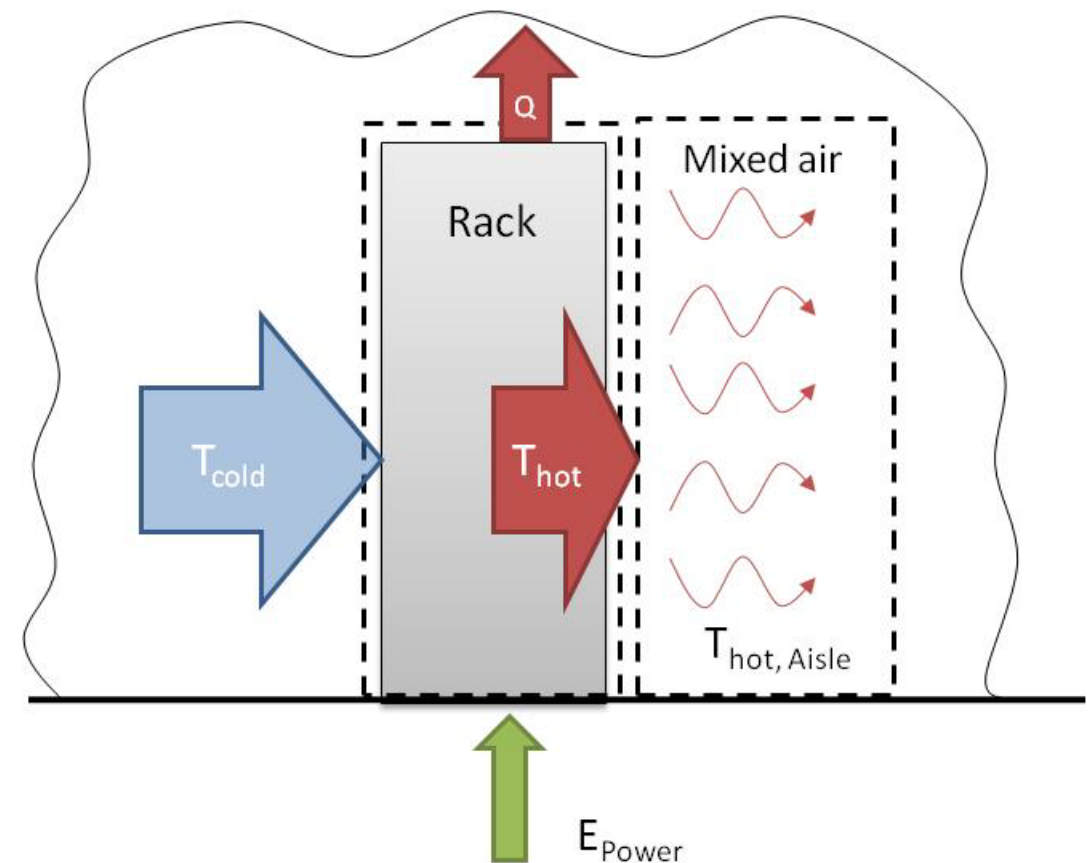


Exergy Destruction Data Center Air Space



Exergy Destruction due to hot aisle mixing

Unrestrained thermal mixing between the hot exhaust air and the hot aisle air is highly irreversible and leads to Exergy destruction



Rack

$$\dot{S}_{gen} \cdot T_{cold} = E_{Power} + \dot{m} \cdot T_{cold} \cdot \left(Cp \cdot \ln \left(\frac{T_{hot}}{T_{cold}} \right) - R \cdot \ln \left(\frac{P_{hot}}{P_{cold}} \right) \right) - \dot{m} \cdot Cp \cdot (T_{hot} - T_{cold})$$

Mixing

$$\dot{S}_{gen} \cdot T_{cold} = \dot{m} \cdot \left[Cv \cdot (T_{hot} - T_{hot,aisle}) + T_{hot} \cdot R - T_{hot,aisle} \cdot R \right] - T_{cold} \cdot \dot{m} \cdot Cp \cdot \ln \left(\frac{T_{hot}}{T_{hot,aisle}} \right)$$



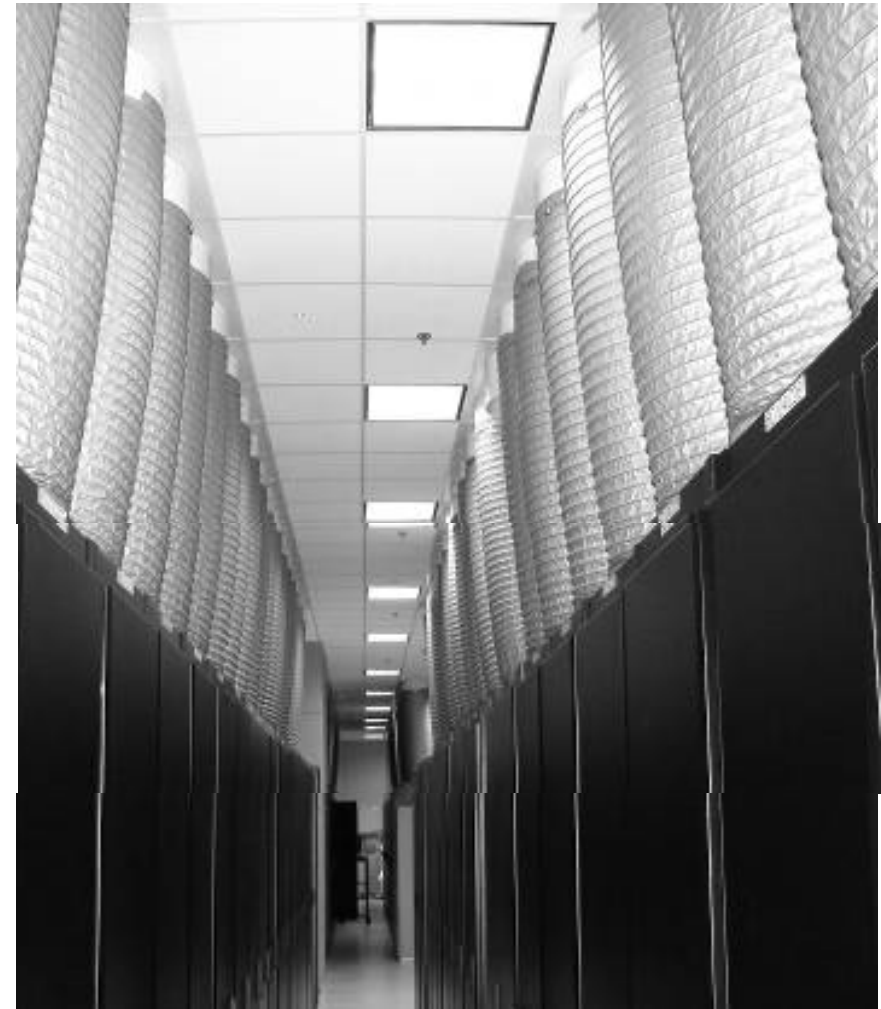
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Reducing hot aisle mixing



Hot aisle containment ADC DC
Sacramento: Courtesy: Wright Line



Direct ducting to return air plenum
Courtesy: ASHRAE

Front End: Combined Heat and Power

Brave New World: CHP used in Data Centers

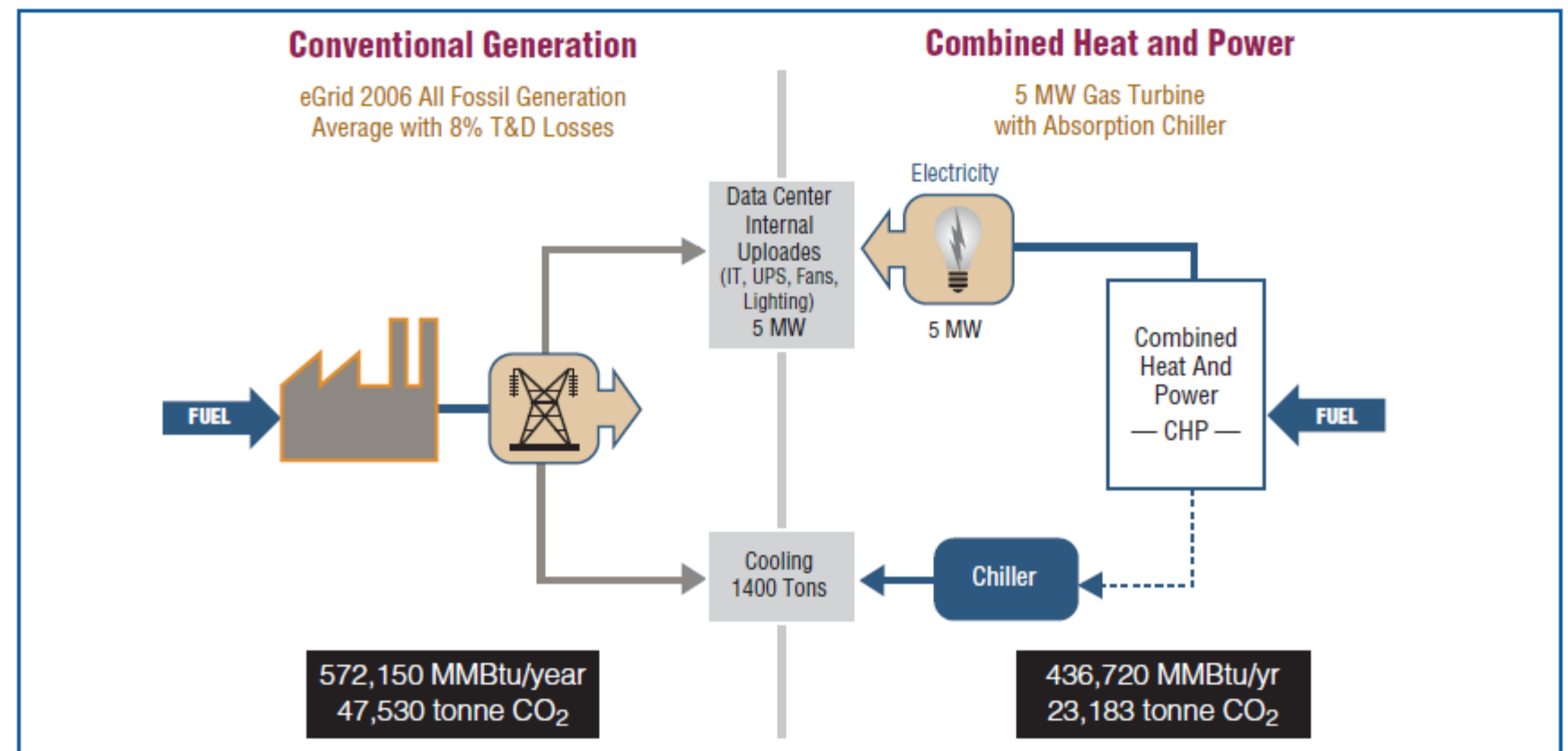
- Qualcomm, San Diego
- Syracuse Univ
- KPMG, NY



Combined Heat and Power

Energy Savings and Energy Reliability for Data Centers

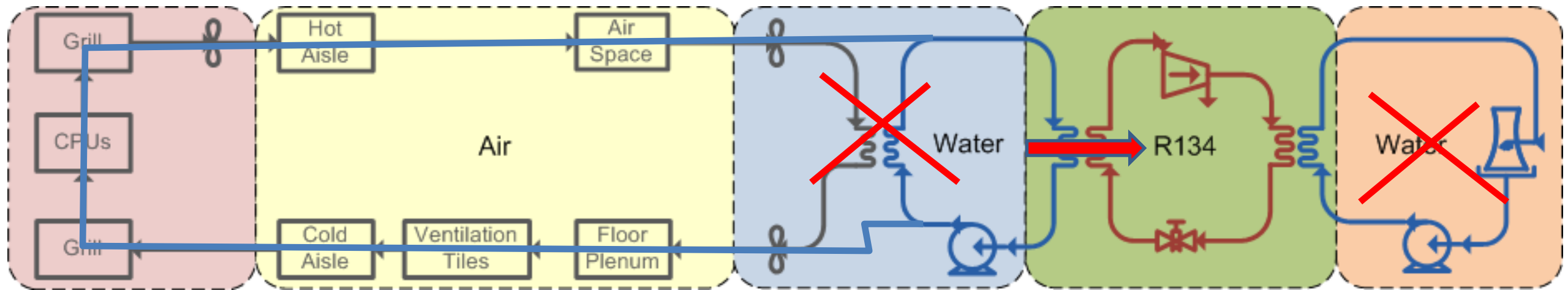
Figure 1: Efficiency and Emissions Benefits of CHP for Data Centers



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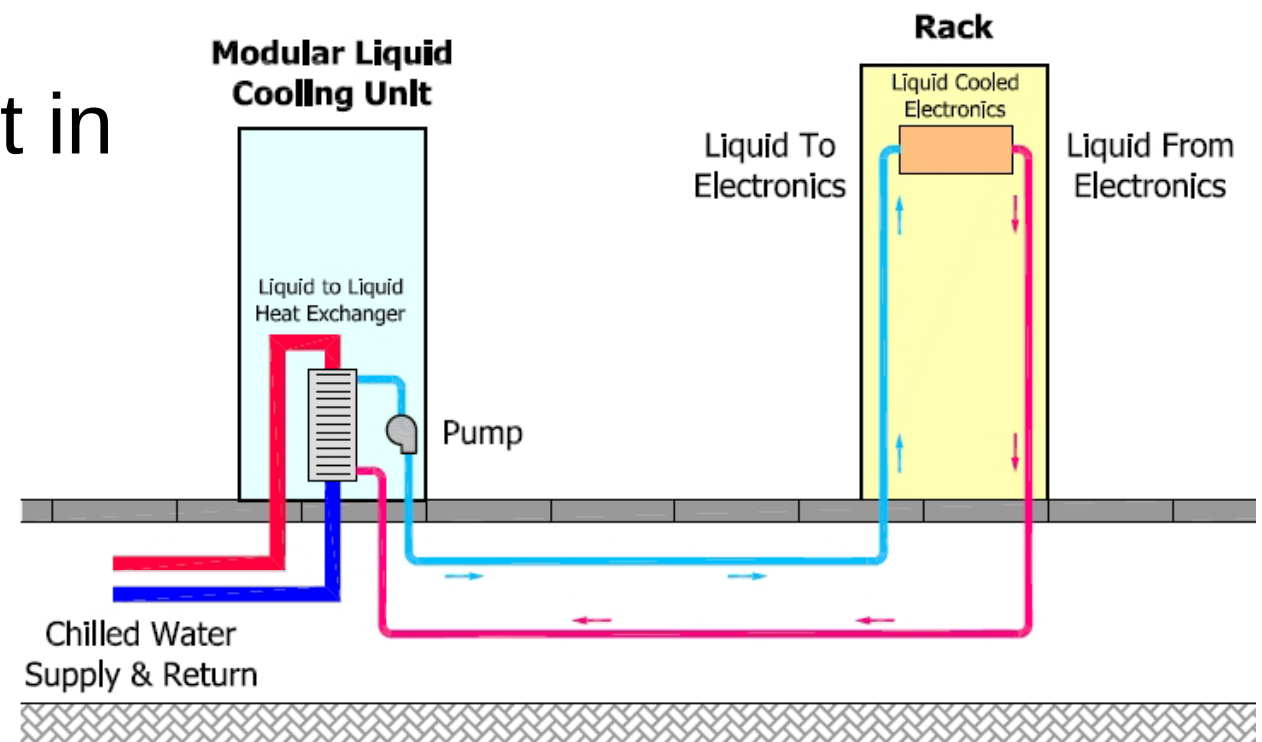
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Back End Waste Energy Re-use



Two compelling reasons to adopt in rack liquid cooling

1. Higher Performance
2. Lower Exergy destruction allowing waste energy re-use



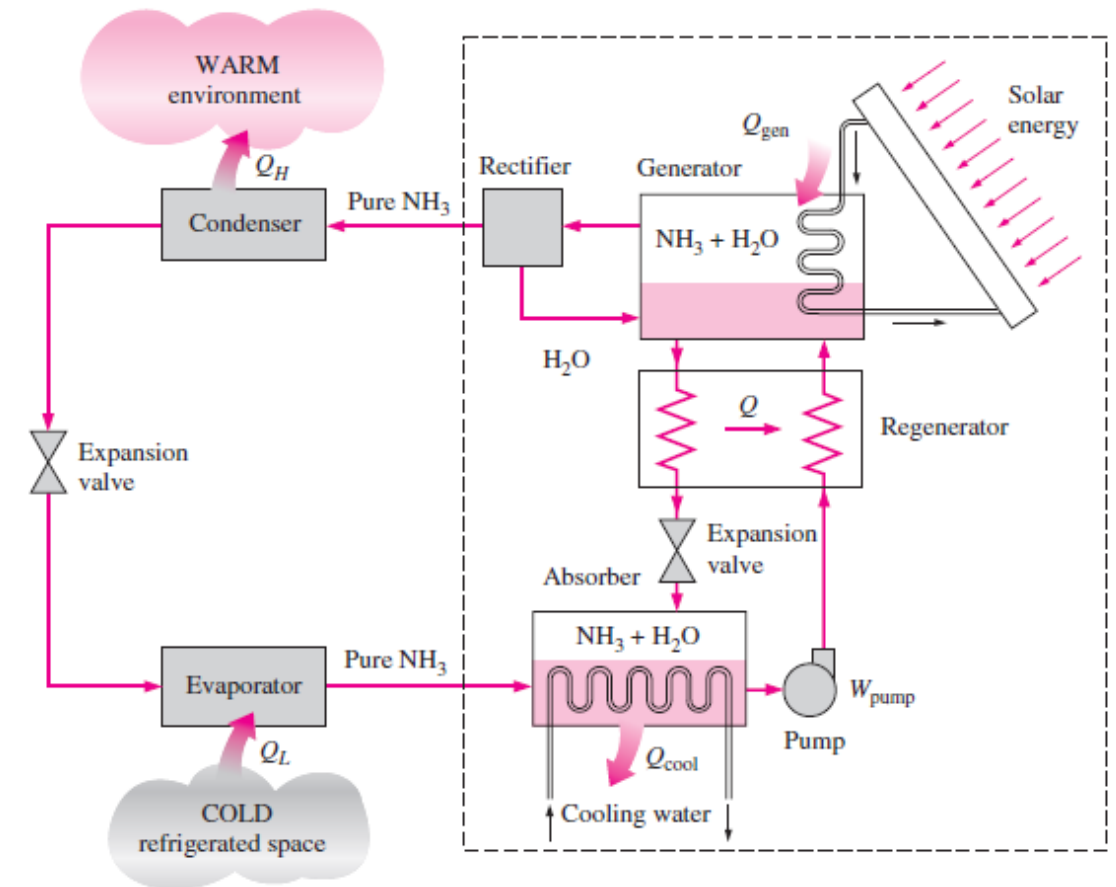
Courtesy: ASHRAE 9.9

WASTE ENERGY RECOVERY:

Heating and Cooling of Habitable Spaces

Absorption Refrigeration

- > Working Fluid: ammonia water ($\text{NH}_3\text{-H}_2\text{O}$) mixture
- > Replace the pump in the refrigeration cycle for a more complex, heat-driven absorption mechanism
- > Temperature range= **70-150°C**



Source: Cengel and Boles 1989, Thermodynamics: an engineering approach



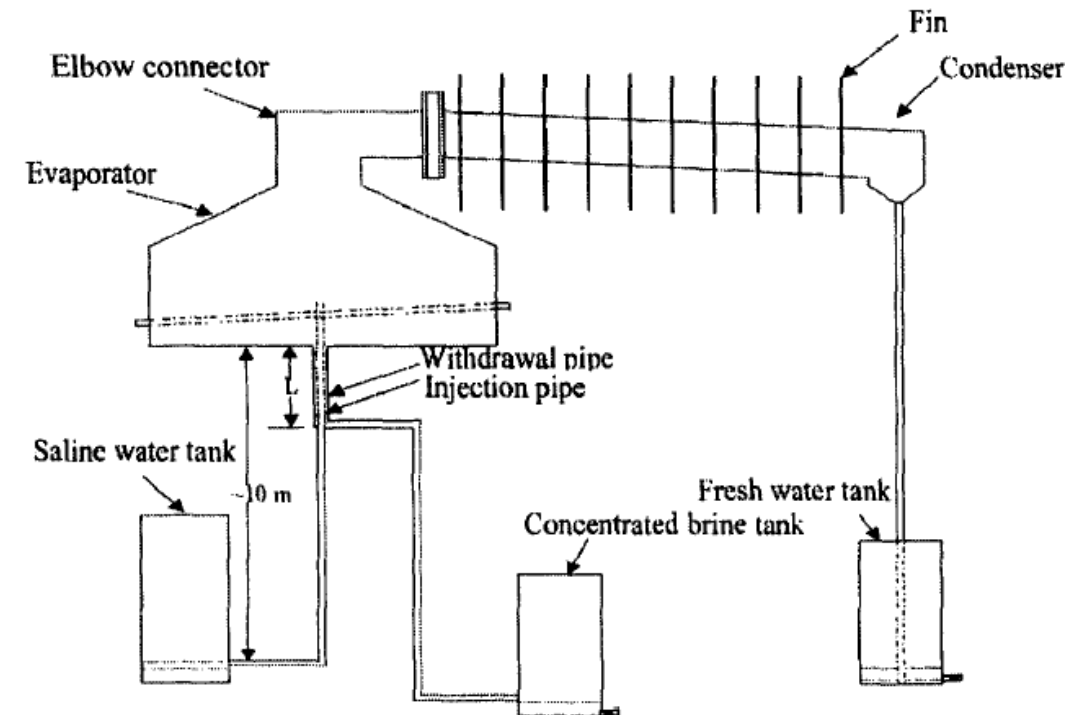
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WASTE ENERGY RECOVERY: Producing Clean Water

Desalination by Humidification- Dehumidification process (HDH)

- > Sea water or brackish water is evaporated and then condensate to separate it from other substances
- > Vacuum can be utilized to evaporated water at much lower temperature than using ambient pressure
- > Temperature range= **40-80°C**



Source: Al-Kharabsheh, Yogi et al. 2003



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CLOSING REMARKS

THERMAL DESIGN DRIVEN BY PERFORMANCE PER UNIT COST OFTEN HAS HAD HIDDEN ENERGY COSTS

AIR COOLED SOLUTIONS ARE BY NATURE PUC DRIVEN; ULTIMATELY DEAD END AT HIGH ENERGY COSTS

PURPOSEFULLY IDENTIFYING IRREVERSIBILITIES LEADS TO MORE ENERGY EFFICIENT DESIGN

NEXT GREAT CHALLENGE IS LEARNING HOW TO DO HOLISTIC SUSTAINABLE DESIGN

